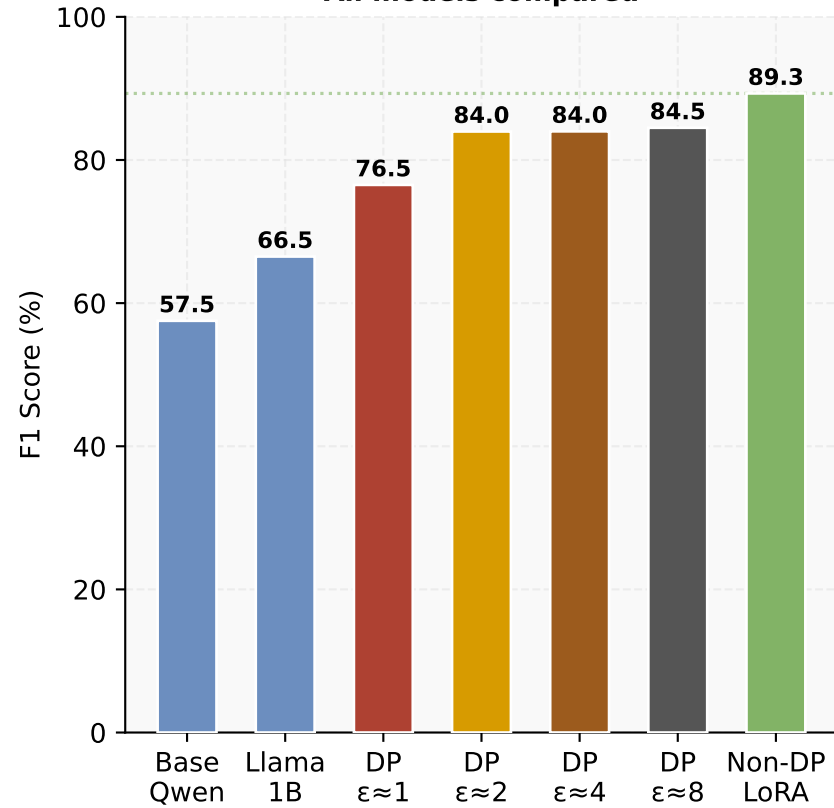


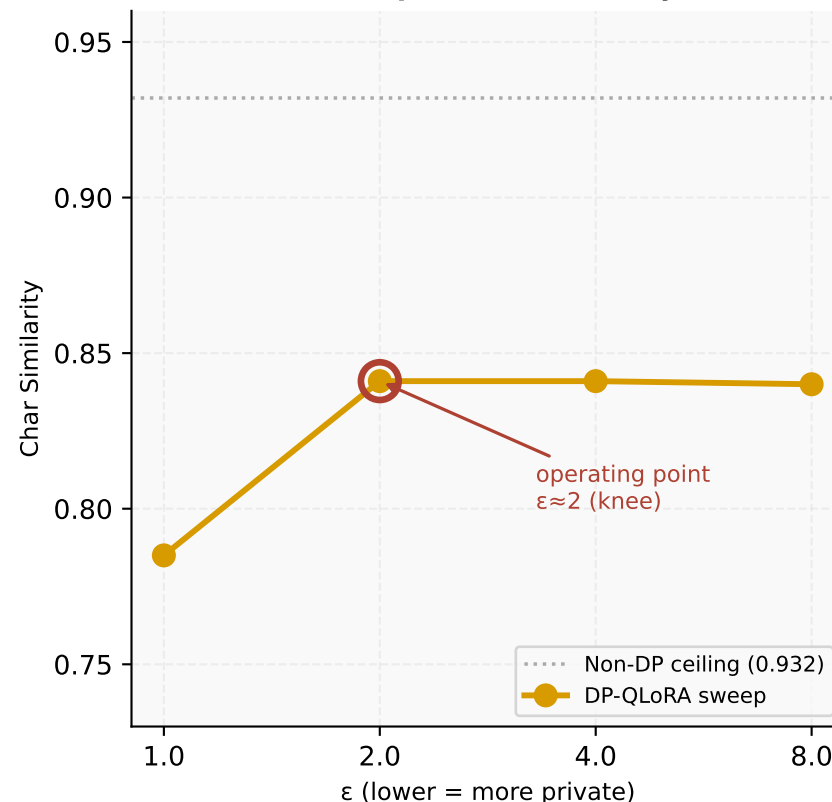
Script 4: Privacy-Utility Tradeoff Dashboard — DP-QLoRA Payee Extraction

$\epsilon=1.9911$, $\delta=2.5\times 10^{-4}$, $C=1.5$, $B=24$, 6 epochs, MAX_LENGTH=64, RDP via Opacus

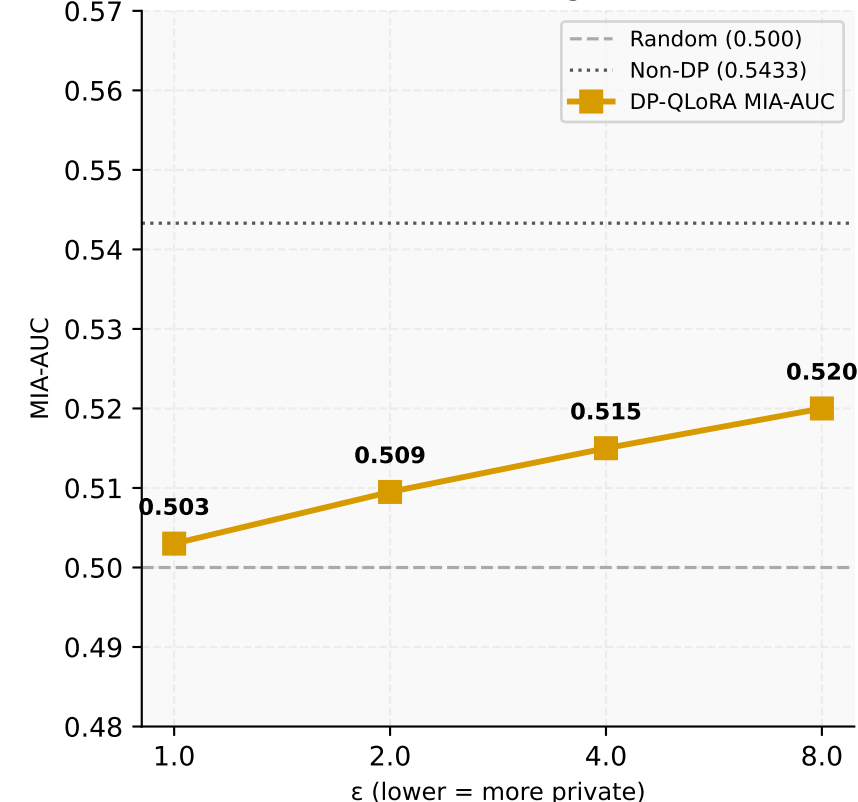
A: Utility vs Privacy Budget
All models compared



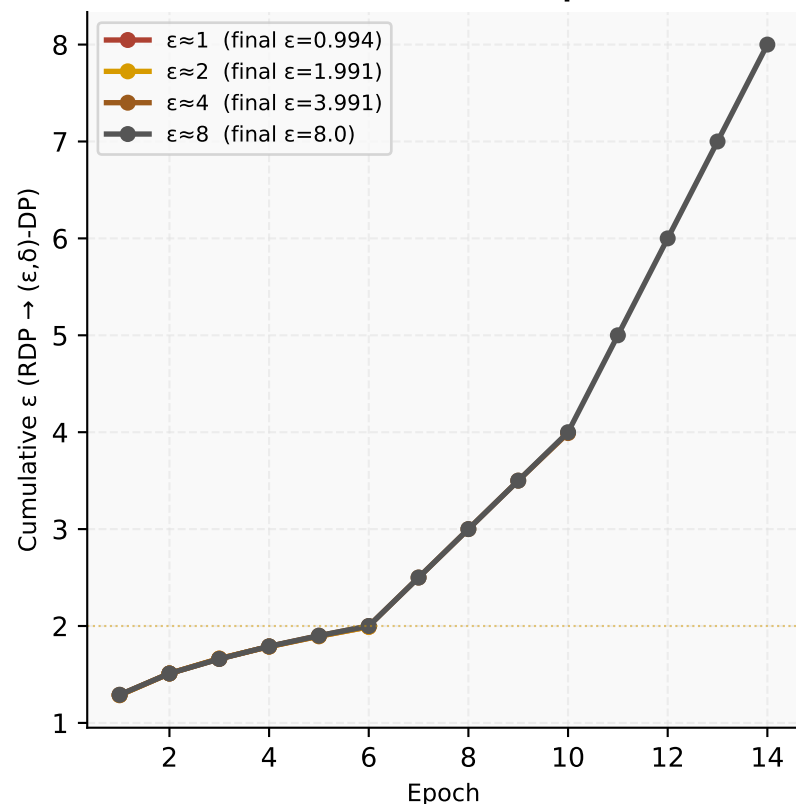
B: Privacy-Utility Knee
 $\epsilon \approx 2$ captures most utility



C: Privacy Floor — MIA-AUC vs ϵ
All near chance $\rightarrow$ formal guarantee holds



D: ϵ Accumulation — All Four DP Runs
RDP accountant via Opacus



E: Full Benchmark Table
All models, all metrics

Model	Type	ϵ	EM (%)	P (%)	R (%)	F1 (%)
Base Qwen 2.5	baseline	—	35.4	55.9	62.1	57.5
Llama-3.2-1B (	baseline	—	55.6	66.3	67.2	66.5
Gemma-2-2B (G	baseline	—	10.7	11.3	11.4	11.3
QLoRA (non-pr	nondp	—	86.2	89.4	89.3	89.3
DP-QLoRA (ϵ	dp	0.994	66.3	73.6	88.1	76.5
DP-QLoRA (ϵ	dp	1.991	62.8	80.4	89.8	84.0
DP-QLoRA (ϵ	dp	3.991	—	—	—	—
DP-QLoRA (ϵ	dp	8.000	—	—	—	—

F: Error Breakdown
DP cost: exact $\rightarrow$ partial, not catastrophic failure

